PROJECT REPORT

Submitted to

DEPARTMENT OF COMPUTER SCIENCE

VIVEKANAND EDUCATION SOCIETY'S

COLLEGE OF ARTS, SCIENCE AND COMMERCE

(AUTONOMOUS), SINDHI SOCIETY, CHEMBUR,

MUMBAI 400071.

GARAGE FINDER PRO!

For Partial Fulfilment for Degree of

Bachelor of Science (Computer Science)

Academic Year (2023-24)

COORDINATOR OF DEPARTMENT

Mr. Kamlakar Bhopatkar

COLLEGE GUIDE

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CERTIFICATE

This is to certify that **Mr. Vikas Ramtirth Jaiswal** of T.Y. B. Sc (Computer Science) affiliated to University of Mumbai has successfully completed a project work entitled.

GARAGE FINDER PRO!

As partial fulfilment of the requirement for the degree of B.Sc. (Computer Science) for the academic year 2023-2024.

COORDINATOR OF DEPARTMENT

Mr. Kamlakar Bhopatkar

COLLEGE GUIDE

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Date:

Examiner:

College Seal

Acknowledgement:

I have great pleasure in presenting this project entitled "**Garage Finder Pro!**" and I grab this opportunity to convey my immense regards towards all the people who with their invaluable contributions made this project successful.

It gives me great pleasure in presenting this project report. Its justification will never complete if I don't express my vote of thanks to our **V.E.S. College** and **Principal Dr. Anita Kanwar**

I sincerely thank and express my profound gratitude to our Project Guides **Mr. Kamlakar Bhopatkar and Dr. Madhavi Vaidya** for timely and prestigious guidance required for the project completion at each phase of the project development.

I also owe to my friends who have been a constant source of help to solve the problems that cropped up during the development of the project, positive criticism, suggestions, constant support, encouragement and guidance towards the successful completion of the project.

SR. NO.	TOPIC	PAGE NO.
	PRELIMINARY INVESTIGATION	
1	Organisational overview	5
2	Description of Existing System.	5
3	Limitations of the Existing system.	6
4	Description Proposed system and its Modules.	7
5	Advantages of Proposed System	9
6	Technologies used	9
	SYSTEM ANALYSIS	
7	Use Case Diagram.	10
8	Activity Diagram.	13
9	NoSQL Schema Design	15
10	Class Diagram	16
	SYSTEM IMPLEMENTATION	
11	Sequence Diagram	18
12	Test Cases	20
13	Component Diagram	26
14	Deployment Diagram	27
15	Images of Application	29
16	Future Enhancements	34
17	References	34

GARAGE FINDER PRO!

Preliminary Investigation

GARAGE FINDER PRO!

Organizational Overview:

Automobile garages are one of the best companions for the drivers on roads for vehicle repair purposes. It helps drivers to repair their vehicles depending upon the vehicle type whether it is a 4-Wheeler, 3-Wheeler, 2-Wheeler or an Electric Vehicle. Some Automobile garages not only repair vehicles of different types but also sell various vehicle spares parts and increasing their business. While there are various ways to find automobile garages, Google Maps has emerged as a popular choice among drivers. With its extensive database and user-friendly interface, drivers can easily locate nearby auto garages and make informed decisions based on ratings and reviews. GoMechanic is also a good option for on-road vehicle repair services. As the number of vehicles on road are increasing day-by-day, the requirements for finding relevant automobile garages are also increasing.

Description of Existing System:

- **Google Maps:**

Google Maps is an integral part of the modern navigation ecosystem used by almost everyone. The existing system Google Maps is a widely popular mapping service that helps users with directions, navigation, location information and various other points of interest including auto garages.

Google Maps primarily relies on user-generated data and business listings to display auto garages. Auto garage owners can create their garage profile by creating their business email id and registering their garage on the Google Maps. Information like name of the garage, address, contact details, operating hours and images can be added to their Google Maps profile. The Information present on Google Maps can be used by users for searching the relevant auto garage. When a user searches for an auto garage, it displays various nearby auto garages and depending on the user's desire, he or she can contact that garage or can directly get directions for that particular garage and can get their vehicles repaired. After a successful repair, the vehicle owner can submit a rating and review for the auto garage which will be publicly shared with the other users using Google Maps. Currently, most of the automobile garage owners have their garages registered on the Google Maps service which helps users to find garages in almost every state of India.

Overall, Google Maps service uses internet and it helps users finding their relevant nearby automobile garages and get their vehicle repaired. It helps users by giving them the contact details and the best routes to reach the garage. Once the vehicle repair is completed, users have the option to submit a rating and review for the auto garage on Google Maps. These reviews provide valuable feedback to other users, helping them make informed decisions when selecting a garage.

- **GoMechanic:**

GoMechanic is a service-based application that helps users to get their vehicles repaired by providing an on-road service. It has their own mechanics and towing services which can be called for service through their application itself as per the requirements. It also provides various vehicle spares parts that user can purchase for their vehicle. Currently, GoMechanic is providing services in almost every state of India. They are also providing their services have in Malaysia also. This widespread availability ensures that users can access their on-road repair services regardless of their location within the country. They also provide various care packages for vehicles which is subscription-based package for the safety of your vehicle. These subscription-based packages provide users with added convenience and peace of mind by including regular maintenance, inspections, and other safety measures for their vehicles. By opting for these care packages, users can ensure the optimal functioning for their vehicles making convenient for them to request assistance.

Limitations of Existing System:

Addressing these limitations is crucial to provide users with a more enhanced and efficient system like "Garage Finder Pro"

- Lack of specific information about the repair capabilities of each garage.
- Incomplete garage profiles with limited details about services offered and pricing information.
- Difficulty in finding garages for specific vehicle types (4-wheelers, 3-wheelers, 2-wheelers, electric vehicles).

- Insufficient information about the expertise of mechanics and availability of parts or equipment.
- Challenges in assessing the suitability and quality of a garage based on available information.
- Limited insights and analytics for both garage owners and users.
- Inefficiency and inconvenience in tracking the progress and status of job cards.

Proposed System:

"Garage Finder Pro" is an innovative app designed to overcome the limitations of current systems and provide best user experience for finding and choosing auto garages. This app focuses on connecting users with garages that have the specific capabilities to repair their vehicle type whether it is a 4-wheeler, 3-wheeler, 2-wheeler or an electric vehicle. The app provides different garage profiles allowing users to view services offered, operating hours, towing service and customer reviews and ratings. This empowers users to make informed decisions ensuring they find the best auto garage for their needs. Users can also access detailed job cards that provide information about the requested or accepted work, estimated costs, and job card status. This transparency allows users to track the progress of their vehicle repairs easily. Additionally, the app offers valuable analytics and insights to garage owners, including metrics such as the number of customers, job cards assigned, in progress, completed, pending and cancelled.

1. Registration and Sign-in:

- User launches the "Garage Finder Pro" app.
- The user is presented with the option to register or sign in if they already have an account.
- Details like user-type (user or garageowner) and mobile number are taken from user.
- The garage owner will also be required to provide more information like services offered, operating hours, towing service availability, etc.
- Garage owners can also upload images of their garage to showcase their facilities and location for the user to reach to their garage.
- The details of the user will get stored in the firestore.
- After successful registration or sign-in, the user is directed to the home screen.

2. Searching and Selecting Garages:

- On the home screen, the user will see a search bar and filters to refine their search.
- The user enters their location or enables location services for automatic detection.
- Based on the entered location the app displays a list of nearby garages.
- The user can apply filters such as vehicle type (4-wheeler, 3-wheeler, 2-wheeler, electric), services offered, operating hours, minimum service cost, etc.
- The user selects a garage from the search results to view its profile.

3. **Viewing Garage Profile:**

- The user is directed to the selected garage's profile page.
- The profile page displays details about the garage, including services offered, operating hours, towing service availability, directions and customer reviews and ratings.
- The user can scroll through customer reviews to get an idea of the garage's reputation.
- If the user wants to proceed with this garage, they can proceed to the next step. Otherwise, they can go back and select a different garage.

4. **Confirming Services and Towing Requirements:**

- The user selects the desired services they require from the garage's profile page.
- If towing service is required, the user can select the option accordingly.
- The location of the vehicle is automatically detected and is available to the towing service man.

5. **Notifications:**

- The app sends notifications to both user and garageowner for various events, including new changes in job card status, customer payments, and new reviews.
- The garageowner will also receive notifications for required towing services, and the user will receive notifications for the same when the garageowner accepts the request.
- The app provides notifications to keep garage owners informed about upcoming requests.

6. **Tracking Repair:**

- After confirming the services and towing requirements, the user can proceed to navigate to garage using garage's location provided in app.
- The user can easily track the progress of their vehicle repairs by accessing the job card and viewing its status (in progress, completed, pending, cancelled).

7. **Job Cards:**

- Garage owners can view all the job cards and can filter them according to their statuses (pending, in progress, cancelled, completed).
- The job card will contain the information about the user, vehicle, requested services and cost, status, etc.

8. **Analytics and Insights:**

- Garage owners can view valuable insights of their garage which will help them to enhance their services.
- Garage owners can get insights about the number of the number of job cards (in progress, completed, pending).

9. **Sharing Reviews:**

- After the repair is completed, the user receives a notification about the status.
- The user can provide a review and rating for the garage based on their experience.
- The review is submitted and added to the garage's profile for other users to see.

Advantages of Proposed System:

Some of the Advantages of the Proposed System:

- Specific capability-based garage search.
- Garage profiles with detailed information.
- Convenient job card management.
- Detailed analytics and insights for garage owners.
- Integration of third-party APIs and libraries, enhances user experience.
- Enhanced user trust and confidence results in retaining customers.

Technologies required for developing the proposed system:

- **Android Studio (Kotlin):** Android Studio is the official Integrated Development Environment (IDE) for Google's Android operating system. Whereas, Kotlin is a High-level programming language. It is one of the official languages of google supported for android application development.

- **Firebase**: Firebase is a comprehensive mobile and web development platform offered by Google. It provides a range of backend services including database management, user authentication, cloud storage, and hosting. Firebase simplifies the development process by offering ready-to-use tools and infrastructure for building high-quality applications.

System Analysis

Use Case

A use case diagram serves as a visual representation of the operations a system undertakes. It comprises key elements such as use cases, actors, and their interconnections. Use cases essentially embody a series of actions that coalesce into a singular unit of work tailored for an actor, who signifies an external user engaging with these operations. In essence, this diagram acts as a pivotal tool for elucidating how a system functions and how external users interact with it, offering a concise yet comprehensive overview of system behaviour.

ELEMENTS OF USE CASE DIAGRAM:

Actor:

Actors in a use case diagram represent entities playing specific roles in a system, typically reflecting real-world user roles. They interact with use cases and should include stakeholders relevant to the system's functionality, with no need to depict entities that don't impact the modeled functions. Actors are symbolized as stick figures outside the system's boundary.

Use Case:

A use case in a diagram visually represents distinct business functions within a system. To identify use cases, begin by listing discrete business functions as basic scenarios. Each function can be categorized as a potential use case. In a use case diagram, use cases are depicted as ellipses, effectively encapsulating the system's specific functionalities. This approach provides a systematic means of representing the core functions the system executes in response to user interactions or external events.

System Boundary:

The system boundary, depicted as a rectangle in a use case diagram, defines the system's scope by encapsulating all its use cases. It distinguishes the system's internal operations from its interactions with the external environment, aiding in effective system communication and design.

Relationships:

The following relationships can be established among use cases:

Extends:

In a use case diagram, one use case can extend another, indicating that the behaviour of the extension use case can be inserted into the extended use case under specific conditions. This relationship is useful for depicting optional functionalities that come after a particular use case. It's represented by a dashed arrow from the optional use case to the base use case, with

the label "«extend»," making it clear that the extended use case can be enhanced or extended with additional features when certain conditions are met.

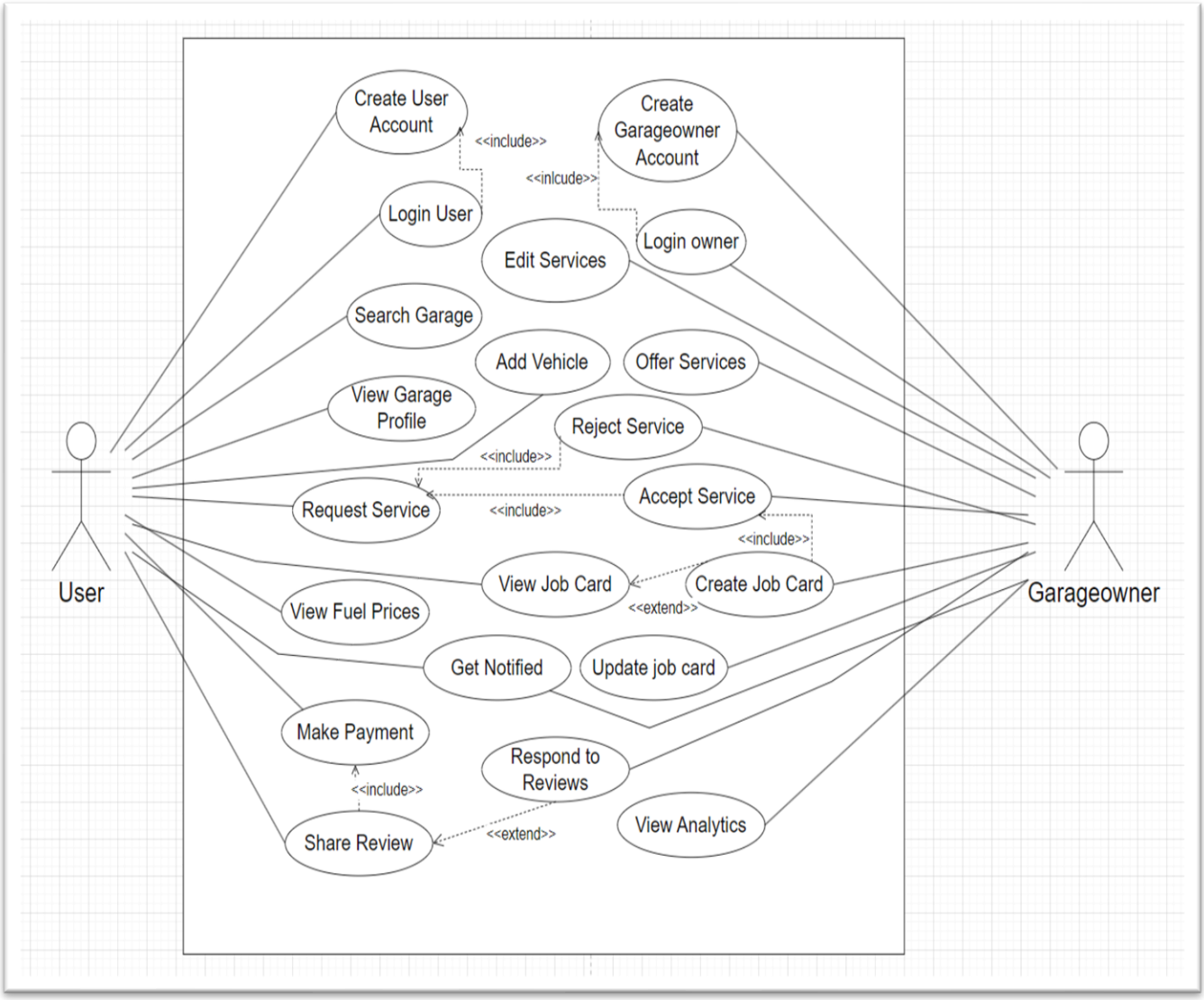
Include:

In a use case diagram, one use case can include another, denoting that the behaviour of the included use case is merged into the behaviour of the including use case. Often, the outcome of the included use case influences the including one. This relationship is represented by a dashed arrow from the base use case to the included use case, labelled "«include»," indicating the integral nature of the included functionality within the encompassing use case.

Basic scenarios for use case

- Registering account
- Logging in
- Adding Vehicle Details
- Adding Garage Details
- Searching Garages
- Sending Request
- Creating Jobcard
- Updating Jobcard
- Reviewing Jobcard
- Making Jobcard Payment
- Viewing Insights

Use Case Diagram:



Activity Diagram

Activity diagrams, akin to flowcharts, depict the sequence of actions or control flow within a system, commonly used in business process modeling and describing use case steps. They feature sequential and concurrent activities, with a defined start (initial state) and end (final state). These diagrams employ symbols for activities, flows, decisions, guards, merge points, time events, and more, enabling comprehensive visualization of intricate processes and system behaviours.

Initial State or Start Point:
In activity diagrams, a small filled circle followed by an arrow symbolizes the initial action state or the starting point for the diagram's activities.

Activity or Action State:
An action state, representing uninterrupted object actions, is portrayed as a rounded rectangle, facilitating visual clarity in diagrams.

Action Flow:
Action flows, known as edges or paths, delineate transitions between action states in diagrams. Typically depicted as arrowed lines, they clarify the sequential progression of activities.

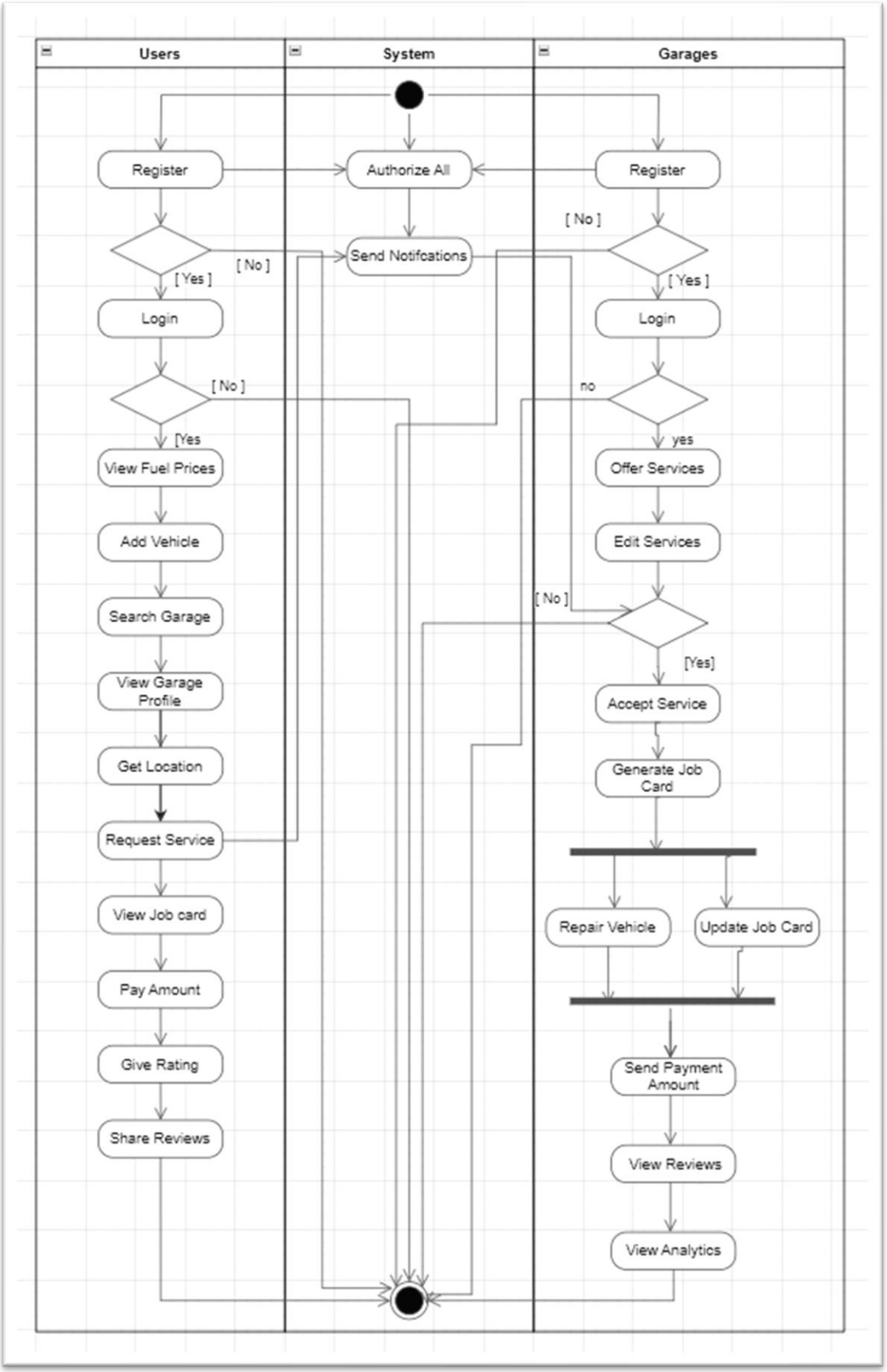
Decisions and Branching:
In Activity diagram, a diamond symbolizes a decision point with multiple paths. To depict activities that hinge on decisions, place a diamond between them. Each outgoing path should be labelled with a condition or guard expression.

Synchronisation:
A fork node in an activity diagram serves to divide a single incoming flow into multiple concurrent flows. It is symbolized as a straight, slightly thicker line, visually indicating the branching of parallel activities.

Swimlanes:
Swim lanes in activity diagrams group related activities into distinct columns, providing a visual organization that clarifies responsibilities or divisions within a process or system. Each lane typically represents a specific entity, role, or department involved in the activities.

Final State or End Point:
In activity diagrams, an arrow directed towards a filled circle enclosed within another circle symbolizes the final action state. This marks the conclusion of an activity or within the diagram's flow, providing clear visual cues for the endpoint.

Activity Diagram



NoSQL Schema Design

In a traditional SQL database, ERD diagrams are used to represent the structure of the data with entities (tables) and their relationships. However, in NoSQL databases, which include document-oriented databases like Firebase Firestore, graph databases, and key-value stores, the data structure is quite different. Therefore, instead of drawing an ERD (which is kind of wrong for Firebase projects because it is non-relational), we can represent the database using boxes.

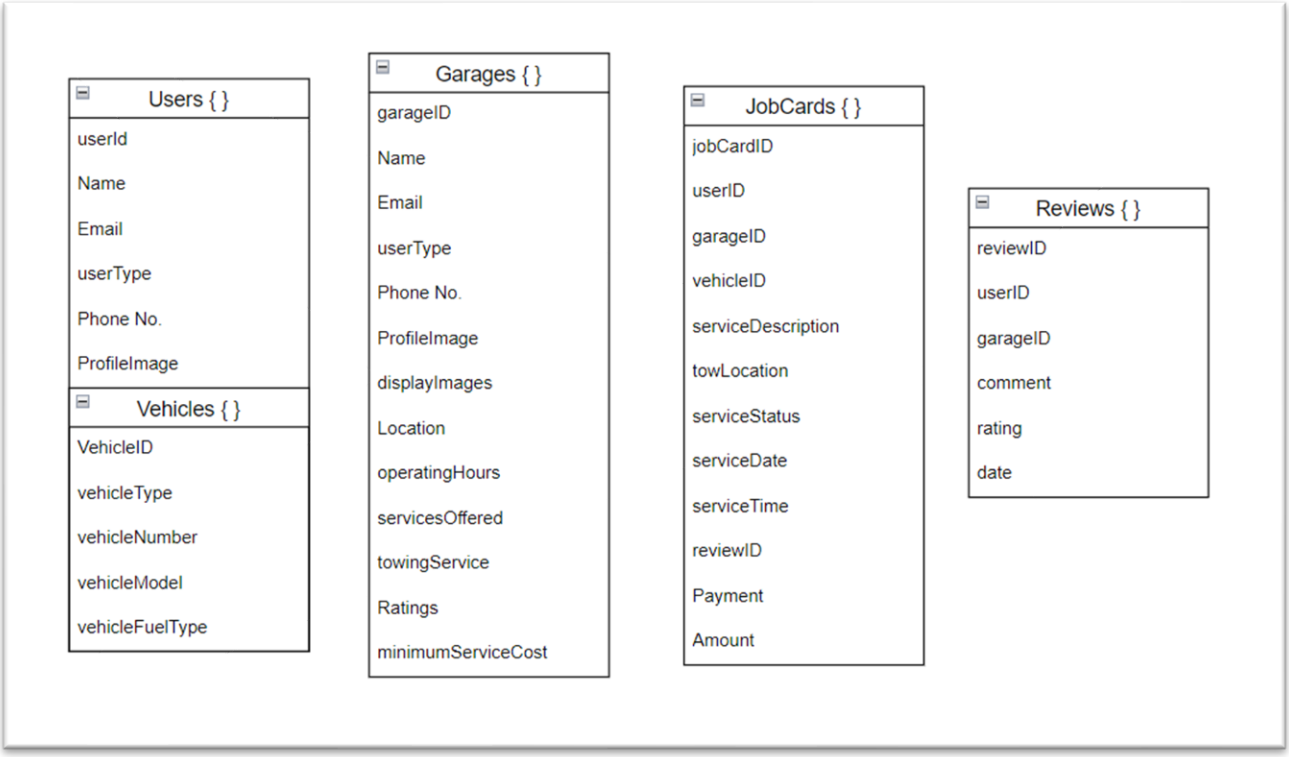
Entities as Collections:

In NoSQL databases, entities are typically organized as collections in document-based databases. These collections serve as containers for related data, similar to how tables represent entities in SQL databases. This flexible schema approach allows for the storage of diverse data structures within each collection, offering adaptability in data modeling.

Attributes as Fields:

In NoSQL databases, entities use fields to store data instead of traditional columns. These fields can accommodate diverse data types, including nested objects and arrays. This flexibility allows for the efficient storage of complex and varied data structures, making NoSQL databases suitable for handling diverse data models and unstructured information.

NoSQL Schema Design



Class Diagram

A class diagram is a fundamental concept in object-oriented modeling and design, often used in software engineering to represent the structure and relationships of classes within a system. Given your background in Android development using Kotlin and Google Firebase, class diagrams can be a useful tool for visualizing the architecture of your Android applications.

Classes:
In a class diagram, classes are represented as rectangles with three compartments. The top compartment typically contains the class name, the middle compartment lists the class attributes (variables), and the bottom compartment includes the class methods (functions).

Attributes:
Class attributes are the properties or variables that belong to a class. They're listed in the middle compartment of the class rectangle.

Methods:
Class methods represent the functions or operations that a class can perform. These are listed in the bottom compartment of the class rectangle.

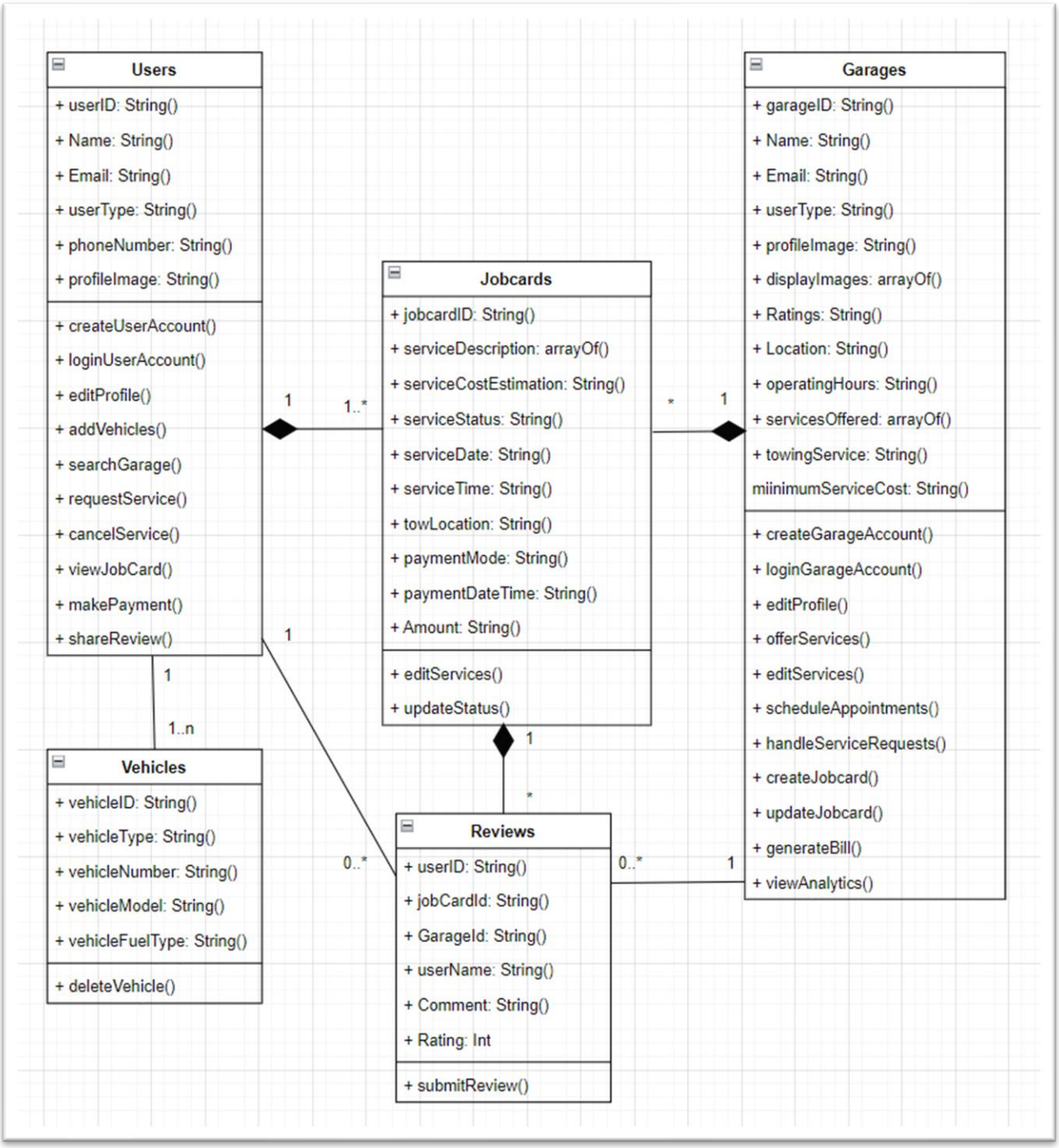
Associations:
In object-oriented programming and UML (Unified Modeling Language), there are several types of associations that represent different kinds of relationships between classes. Here are some common types of associations:

Association:
This is a basic relationship between classes, indicating that one class is associated with another in some way. It's a generic term for any connection between classes.

Aggregation:
Aggregation represents a "whole-part" relationship, where one class (the whole) contains or is composed of other classes (the parts). It's typically depicted as a diamond shape.

Composition:
Composition is a stronger form of aggregation, indicating that the parts are essential to the whole and can't exist independently. It's also depicted with a diamond shape.

Class Diagram



Sequence Diagram

A sequence diagram is a visual representation in software engineering that depicts the interactions between different components or objects within a system over a specific period. It illustrates the flow of messages or actions between these entities, helping developers understand the order and timing of these interactions. Sequence diagrams are commonly used to model and document the behaviour of software systems, making them a valuable tool in the design and analysis of complex systems.

In short, A sequence diagram is a UML (Unified Modeling Language) diagram that provides a detailed view of how various components or objects in a software system interact with each other.

Participants:
In a sequence diagram, participants are represented by lifelines, typically depicted as vertical lines. These lifelines represent objects, classes, or components involved in the interaction. Messages flow between these lifelines to represent interactions.

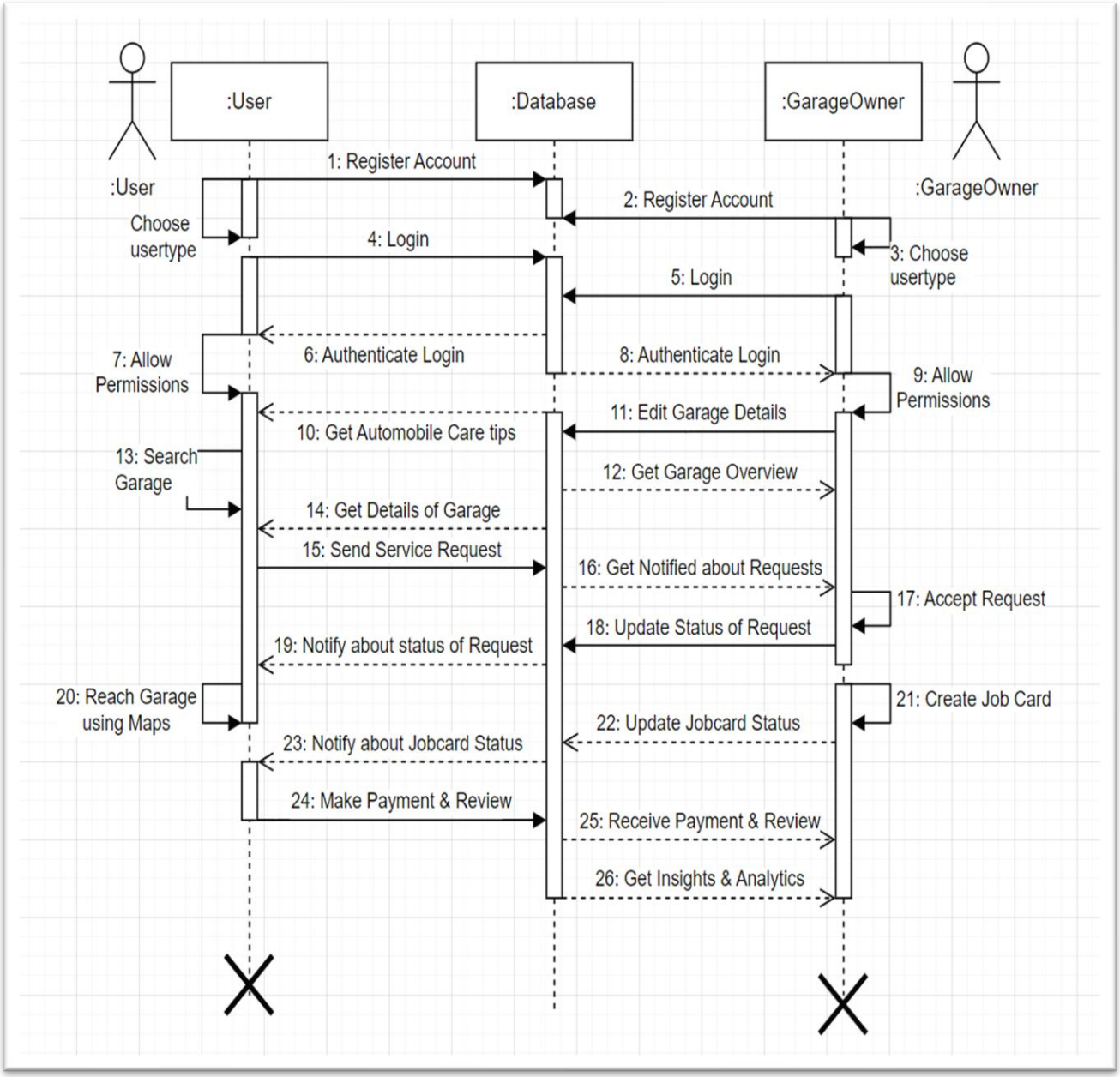
Messages:
Messages are depicted as arrows that flow between lifelines. They represent communication or interactions between objects. Messages can be synchronous, asynchronous, or create/destroy messages, depending on the type of interaction.

Activation Bar:
An activation bar is a horizontal bar along a lifeline that shows the period during which an object is active and processing a message. It helps visualize the timing of interactions.

Focus on Time:
Sequence diagrams are especially useful for emphasizing the chronological order of interactions. Developers can see how messages are exchanged over time, aiding in understanding system behaviour and potential bottlenecks.

Lifelines:
Lifelines visually represent the lifespan or existence of the associated object or component within the sequence diagram. The top of the lifeline represents the creation of the object, and the bottom represents its destruction, if applicable.

Sequence Diagram



Test Cases

TestCaseID	Module	Form	Test Conditions	Steps	Input Test Data	Expected Output	Actual Output	Status
TC1	Registration	Registration Form	Provide valid Details by following all validations	1. Click on Sign-up Button 2. Select User Type 3. Fill Sign-up Details	1. User Type 2. Name 3. Email 4. Password 5. Re-Password 6. Accept T&Cs	1. Registered Successfully 2. Verify Email Address	1. Registered Successfully 2. Verify Email Address	Pass
TC2	Login	Sign-in Form	Enter Valid Credentials	1. Click on Sign-in Button	1. Enter Email 2. Enter Password	Login Successful	Login Successful	Pass
TC3	Login	Sign-in Form	Enter Invalid Credentials	1. Enter Invalid Credentials	1. Enter Invalid Credentials	Invalid Credentials	Invalid Credentials	Pass
TC4	Google Sign-in	Sign-in Form	Select Valid Google Email ID	1. Select Google Account	1. Select Google Account	Login Successful	Login Successful	Pass
TC5	Forgot Password	Sign-in Form	Enter Valid Email	1. Enter Valid Email	1. Enter Valid Email	Password Reset	Password Reset	Pass
TC6	Forgot Password	Sign-in Form	Enter Unregistered Email	1. Enter Unregistered Email	1. Enter Unregistered Email	Email is not registered	Email is not registered	Pass
TC7	Login	Sign-in Form	User should be already Signed in	1. Sign in and Restart the App	Not Applicable	Directly Signs you in, no need for re-login	Directly Signs you in, no need for re-login	Pass
TC8	User	Home Page	Allow Location Permissions & Turn on Location Services	1. Allow Location Permissions 2. Turn on GPS Service	Not Applicable	Location of user	Location of user	Pass
TC9	User	Home Page	Allow Location Permissions & Turn on Location Services	1. Allow Location Permissions 2. Turn on GPS Service	Not Applicable	Local Fuel Prices of Petrol, Diesel & CNG	Local Fuel Prices of Petrol, Diesel & CNG	Pass
TC10	All	Offline Page	No Internet Connection / Device connected to	1. Turn off wifi/data connection	Not Applicable	Offline Page Appears.	Offline Page Appears	Pass

			WIFI but no internet					
TC11	User	Profile Page	Click on Sign out Button	1. Go to Profile Page 2. Click on Sign out Button	Not Applicable	Return to Sign in Page	Return to Sign in Page	Pass
TC12	User	Profile Page	Open Profile Page & Internet Connection	1. Go to Profile Page	Not Applicable	Displays Details of the user	Displays Details of the user	Pass
TC13	User	Profile Page	Add Vehicle	1. Open Profile Page 2. Click on add Vehicle Button 3. Fill Details and Submit	1. Vehicle Name 2. Vehicle Number 3. Vehicle Model 4. Vehicle Fuel Type 5. Vehicle Type	Vehicle Added Successfully and Displays on Profile	Vehicle Added Successfully and Displays on Profile	Pass
TC14	User	Profile Page	Add Contact	1. Open Profile Page 2. Click on add Contact 3. Fill Contact and Submit	1. Contact Number	Contact Number Added Successfully	Contact Number Added Successfully	Pass
TC15	Garage	Garage Profile Page	Add Garage Details	1. Open Profile Page 2. Click on Edit Profile 3. Add details and upload profile Image	1. Garage Name 2. Operating hours 3. Towing Service 4. Minimum service cost 5. Profile Image	Profile Updated Successfully	Profile Updated Successfully	Pass
TC16	Garage	Garage Profile Page	Add Garage Showcase Image	1. Click on Edit Showcase Images Icon	1. Select Images from gallery	Showcase Images Added Successfully and Reflecting on Profile	Showcase Images Added Successfully and Reflecting on Profile	Pass
TC17	Garage	Garage Profile Page	Open Profile Page & Internet Connection	1. Open Profile Page	Not Applicable	Current Rating & Recent Reviews will be displayed on Profile	Current Rating & Recent Reviews will be displayed on Profile	Pass

TC18	Garage	Garage Home Page	Open Home Page of the Garage Profile	1. Sign in as a Garage Owner	Not Applicable	Displays All the current Jobcards under Garage and Earnings Insight	Displays All the current Jobcards under Garage and Earnings insight	Pass
TC19	User	Garage Search Page	Open Garage Search Page	1. Open garage Search Page	Not Applicable	Displays Different Garages Details	Displays Different Garages Details	Pass
TC20	User	Garage Search Page	Open Garage Search Page and Click on More Details of any garage	1. Open garage Search Page 2. Click on More Details	Not Applicable	Displays a Detailed Profile of that Garage	Displays a Detailed Profile Of that Garage	Pass
TC21	User	Garage Search Page	Open Garage Search Page, click on any garage's More Details and Click on Contact	1. Open garage Search Page 2. Click on More Details of Any garage 3. Click on Contact	Not Applicable	Contacts Garages on their stored numbers	Contact Garages on their stored numbers	Pass
TC22	User	Garage Search Page	Open Garage Search Page, click on any garage's More Details and Click on Google Maps under Directions	1. Open garage Search Page 2. Click on More Details of Any garage 3. Click on Google Maps	Not Applicable	User gets the Precise Location of the garage and can navigate to it	User gets the Precise Location of the garage and can navigate to it	Pass
TC23	User	Garage Search Page	Open garage Search page and Click on Request Service from a garage	1. Open garage Search page 2. Click on Request Service 3. Fill Details and Request Submit	1. Service Name 2. Vehicle Details 3. Tow Service	Respective Garage Owner gets Request Notification and can View it	Respective Garage Owner gets Request Notification and can view it	Pass
TC24	Garage	Garage Request Process panel	Open garage request process panel and change the dropdown to new requests	1. Open Garage Request Process panel 2. Check new requests and accept it	1. Select new Requests from Drop Down	Request gets Accepted and the respective user gets notified	Request gets Accepted and the respective user gets notified	Pass

TC25	Garage	Garage Request Process panel	Open garage request process panel and update the jobcard status	1. Open Garage Request Process panel 2. Go to relevant jobcard and click on view more 3. Select status type from dropdown 4. Click on Update job card	1. Click on View More 2.Select relevant jobcard update 3. Click update jobcard	Job card gets Updated and the respective user gets notified	Job card gets Updated and the respective user gets notified	Pass
TC26	User	User Request Process Panel	Open user request process panel and check completed jobcards and make payments	1. Open user request process panel 2. Select Completed Jobcards from dropdown 3. Select a relevant job card and view more 4. Make Payments Accordingly	1. Click on View More 2.Select relevant jobcard Completed 3. Click on Make Payment	Payment Successful and the respective garage owner will be notified about it	Payment Successful and the respective garage owner will be notified about it	Pass
TC27	User	User Request Process Panel	Open user request process panel and check completed jobcards and click on leave a review	1. Open user request process panel 2. Select Completed Jobcards from dropdown 3. Select a relevant job card and view more 4. Make Ratings and Reviews Accordingly	1. Rating 2.Review	Rating and Review gets submitted and reflects on the respective garage owners' profile	Rating and Review gets submitted and reflects on the respective garage owners' profile	Pass

TC28	Garage	Garage Profile Page	Open garage Profile page and click on edit banner image icon	1. Sign in as a Garage Owner 2. Open Profile Page 3. Click edit Banner image icon 4. Select Camera or gallery 5. Click on OK and banner gets updated	1. Select Relevant Image	Garage Profile Banner Image Gets Updated	Garage Profile Banner Image Gets Updated	Pass
TC29	Garage	Garage Profile Page	Open garage Profile page and click on edit showcase image icon	1. Sign in as a Garage Owner 2. Open Profile Page 3. Click edit showcase image icon 4. Select Max 7 Images from gallery 5. Click on OK and showcase images gets updated	1. Select Max 7 Relevant Images from Gallery	Garage Profile showcase Images Gets Updated	Garage Profile showcase Images Gets Updated	Pass
TC30	Garage	Garage Profile Page	Open garage Profile Page and click on edit specialization icon	1. Sign in as a Garage Owner 2. Open Profile Page 3. Click on edit Specialization 4. Add your Top 5 Specialization using dropdowns 5. Click on Update Specialization button	1. Top 5 garage Specialization using drop down menus	Garage Specializations gets updated and reflects on the respective profile	Garage Specializations gets updated and reflects on the respective profile	Pass

TC31	User	Profile Page	Open Profile Page and click on delete a vehicle icon and click on yes on alert box	1. Open user profile page 2. Click on detail vehicle icon 3. Click on yes on alert box	Not Applicable	The Respective Vehicle gets Deleted from user profile	The Respective Vehicles gets deleted from user profile	Pass
TC32	Garage	Garage Home Page	Open garage home page and switch on the garage status to make its status as Open	1. Open Garage home Page 2. Switch the Status by clicking on it	Not Applicable	The respective garage gets status as Open and now is visible to other users	The respective garage gets status as Open and now is visible to other users	Pass
TC33	Garage	Garage Insights Page	Open garage Insights page	1. Open garage insights page	Not Applicable	Relevant Visualized graphs will be displayed	Relevant Visualized graphs will be displayed	Pass
TC34	User	Garage Search Page	Open garage search and request a service and check on the Tow Required	1. Open garage search page 2. request a service by filling details and check tow required	1. Service Name 2. Vehicle Details 3. Tow Service	The precise location of the user will be shared with the requested garage owner	The precise location of the user will be shared with the requested garage owner	Pass

Component Diagram

A component diagram is a type of UML (Unified Modeling Language) diagram used in software engineering to visualize the high-level structure and organization of a software system or application. While it may not be directly related to your Android development and Firebase expertise, understanding component diagrams can still be valuable in the context of software design and architecture.

A component diagram is primarily focused on depicting the various components or modular units that make up a software system and how they interact with each other. These components can represent classes, packages, libraries, or even physical modules, depending on the level of abstraction you want to depict.

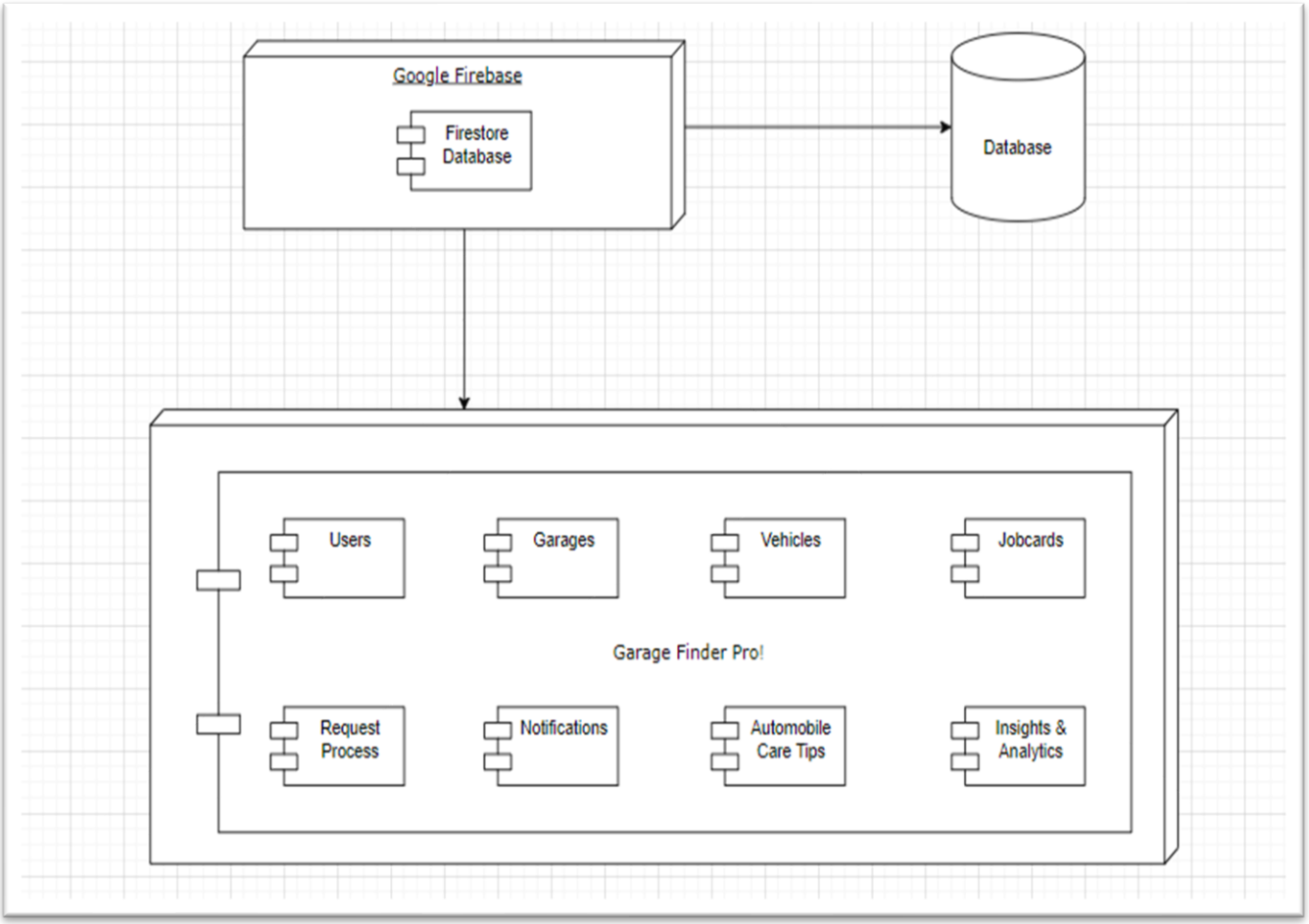
Components:
Components are the fundamental building blocks of a system. They encapsulate a set of related functionalities and can be represented as rectangles in the diagram. Each component typically corresponds to a software module or a class.

Composite Components:
In larger systems, it's common to group related components into higher-level composite components. These composite components can be represented as larger rectangles containing other components, showing a higher level of abstraction.

Database Component:
Represent the database as a separate component within your component diagram. This component should have a name that clearly identifies it as the database, such as "Database" or the name of the specific database system you are using, like "MySQL," "Firebase Firestore", etc.

Connectors:
Connectors are used to depict the relationships between components and interfaces. They define how data or control flows between components.

Component Diagram



Deployment Diagram

A deployment diagram is a type of diagram in the Unified Modeling Language (UML) that is used to visualize and describe the architecture of a system as it is deployed in a real-world environment. Deployment diagrams can be valuable for understanding how your mobile app interacts with backend services, servers, and other components when it's deployed in a production environment.

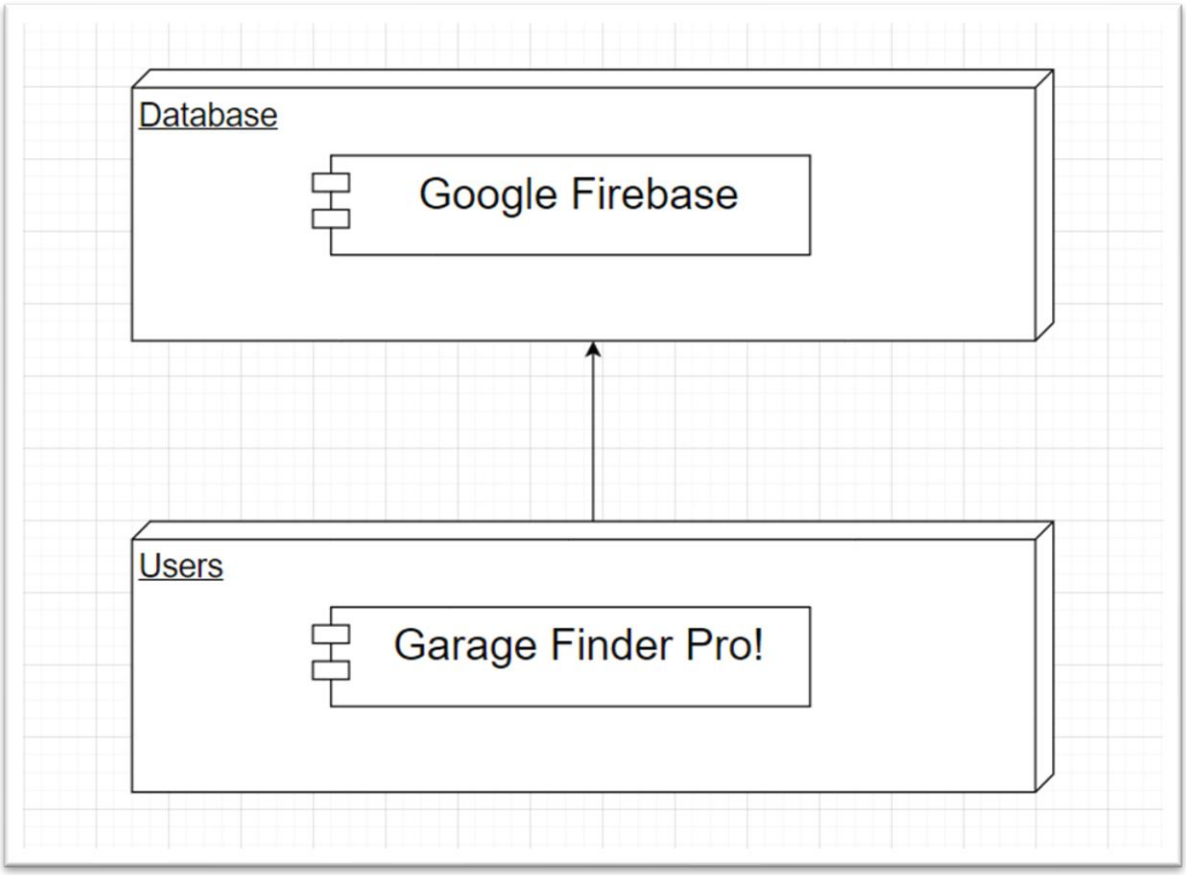
Nodes:

Nodes represent physical or virtual computational resources in your deployment. In your case, these could be servers, cloud services, or even mobile devices. For example, you might have a node for your Android devices and another for the Firebase server.

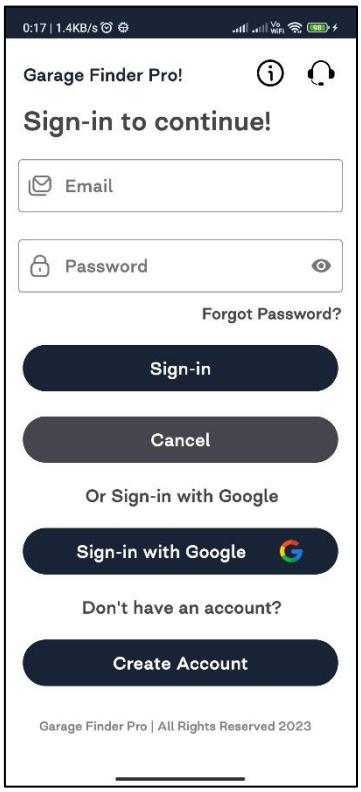
Dashed Lines:

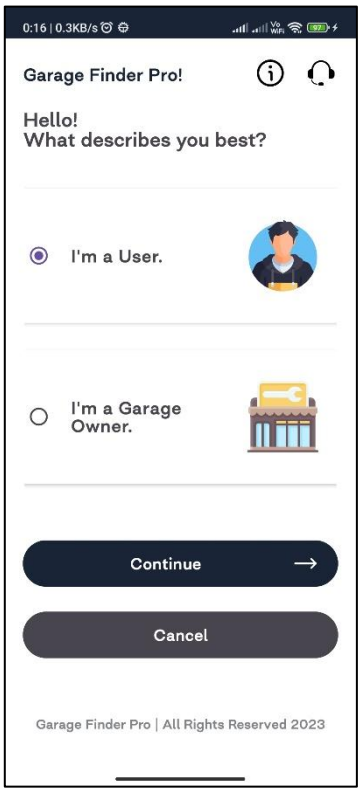
The dependency on node on components is depicted using dashed lines. This means that a node uses the services of the components that are executing on another node

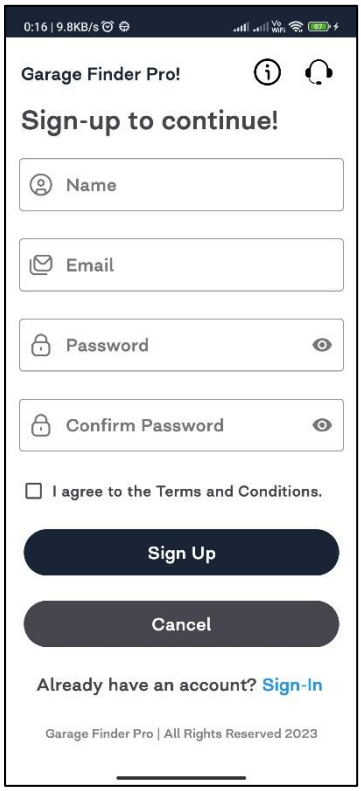
Deployment Diagram

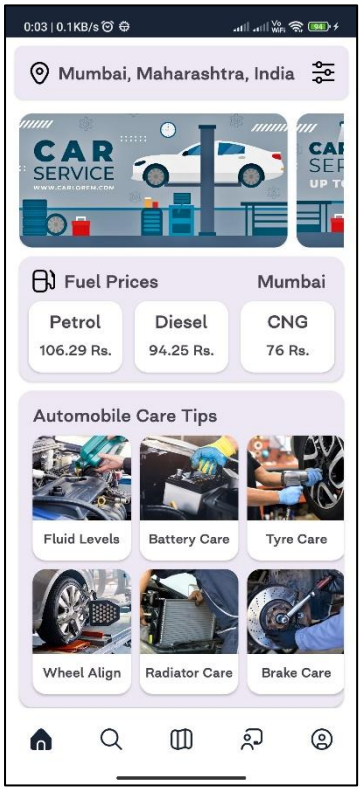


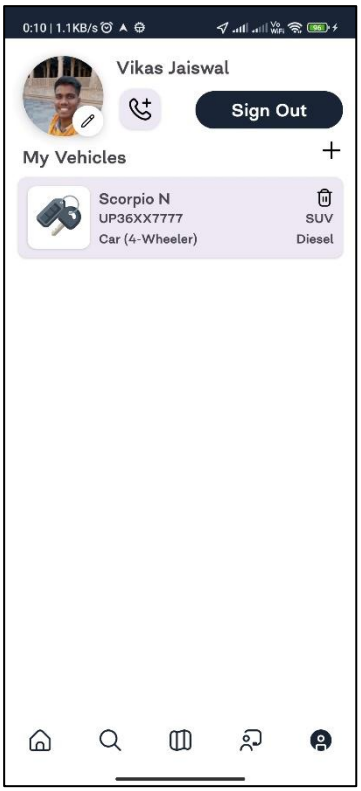
Images of Application

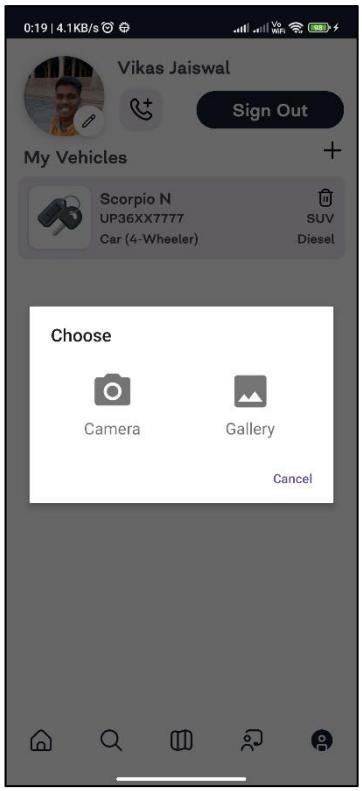


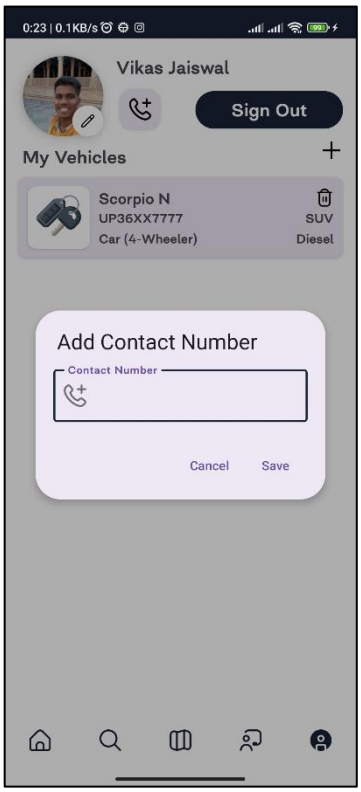


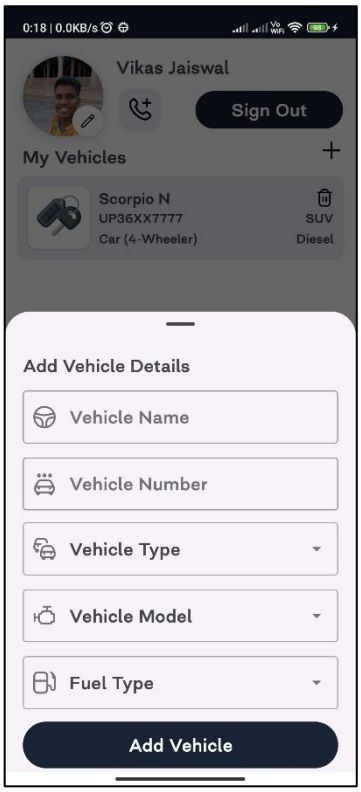


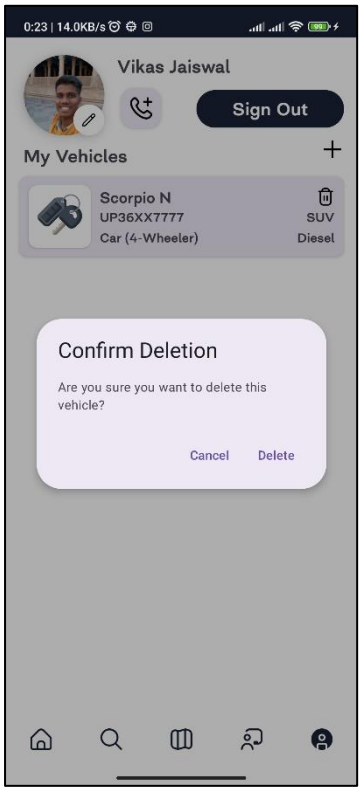


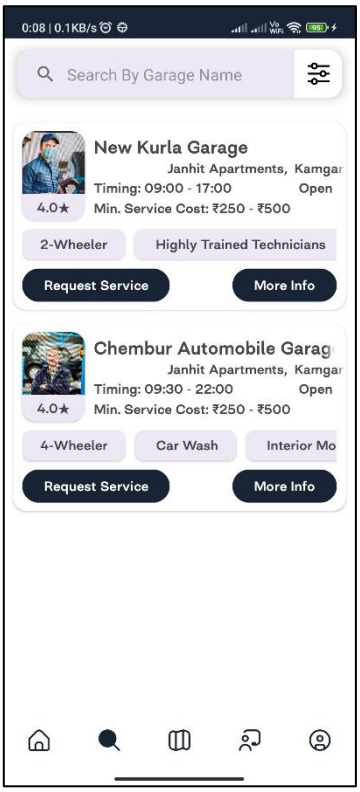


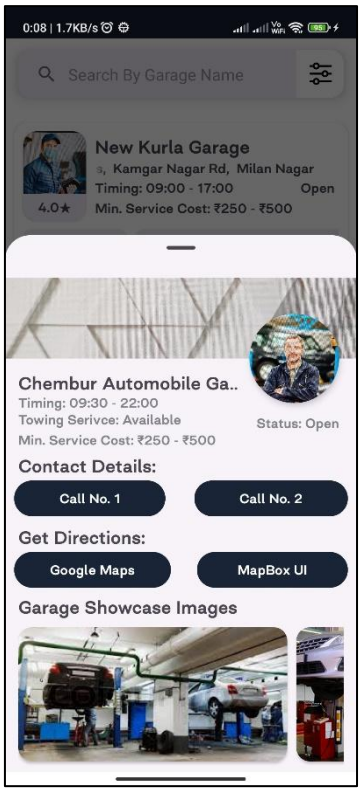

0:19 | 4.1KB/s
Vikas Jaiswal
Sign Out
My Vehicles
Scorpio N
UP36XX7777
Car (4-Wheeler)
SUV
Diesel
Choose
Camera
Gallery
Cancel


0:23 | 0.1KB/s
Vikas Jaiswal
Sign Out
My Vehicles
Scorpio N
UP36XX7777
Car (4-Wheeler)
SUV
Diesel
Add Contact Number
Contact Number
Cancel
Save

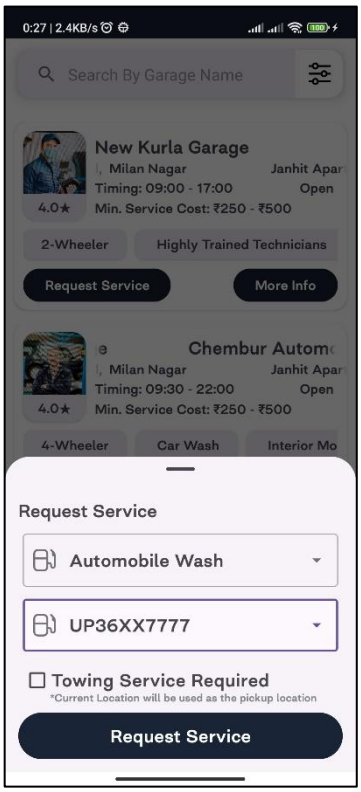

0:18 | 0.0KB/s
Vikas Jaiswal
Sign Out
My Vehicles
Scorpio N
UP36XX7777
Car (4-Wheeler)
SUV
Diesel
Add Vehicle Details
Vehicle Name
Vehicle Number
Vehicle Type
Vehicle Model
Fuel Type
Add Vehicle

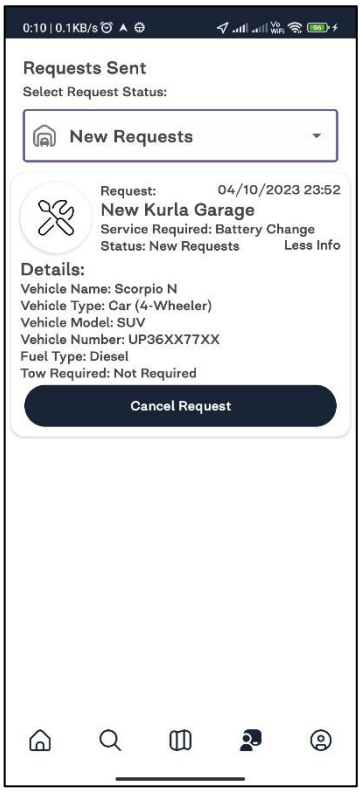

0:23 | 14.0KB/s
Vikas Jaiswal
Sign Out
My Vehicles
Scorpio N
UP36XX7777
Car (4-Wheeler)
SUV
Diesel
Confirm Deletion
Are you sure you want to delete this vehicle?
Cancel
Delete

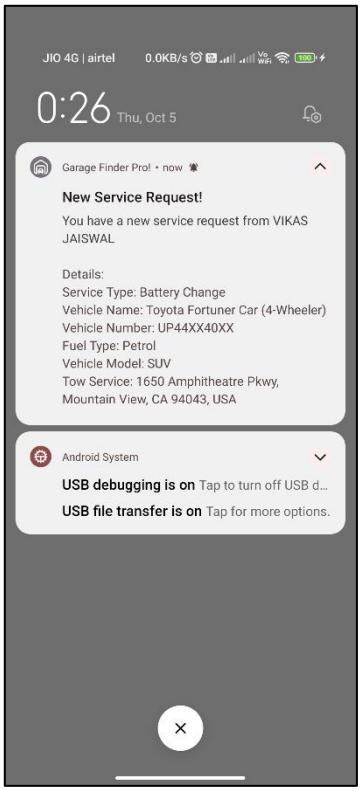

0:08 | 0.1KB/s
Search By Garage Name
New Kurla Garage
Janhit Apartments, Kamgar
Timing: 09:00 - 17:00
Open
4.0★
Min. Service Cost: ₹250 - ₹500
2-Wheeler
Highly Trained Technicians
Request Service
More Info
Chembur Automobile Garag
Janhit Apartments, Kamgar
Timing: 09:30 - 22:00
Open
4.0★
Min. Service Cost: ₹250 - ₹500
4-Wheeler
Car Wash
Interior Mo
Request Service
More Info

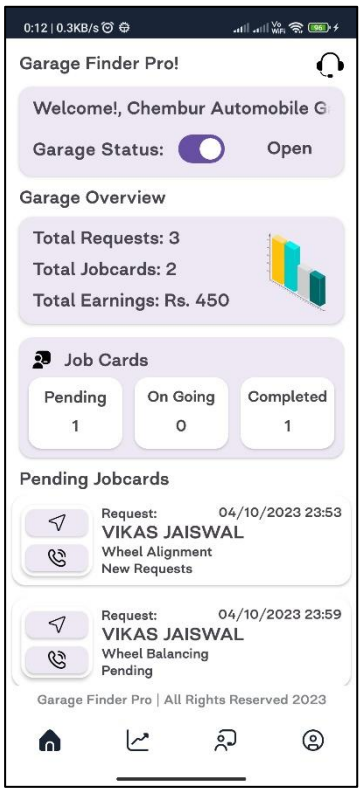

0:08 | 1.7KB/s
Search By Garage Name
New Kurla Garage
Kamgar Nagar Rd, Milan Nagar
Timing: 09:00 - 17:00
Open
4.0★
Min. Service Cost: ₹250 - ₹500
Chembur Automobile Ga..
Timing: 09:30 - 22:00
Towing Serivce: Available
Min. Service Cost: ₹250 - ₹500
Status: Open
Contact Details:
Call No. 1
Call No. 2
Get Directions:
Google Maps
MapBox UI
Garage Showcase Images

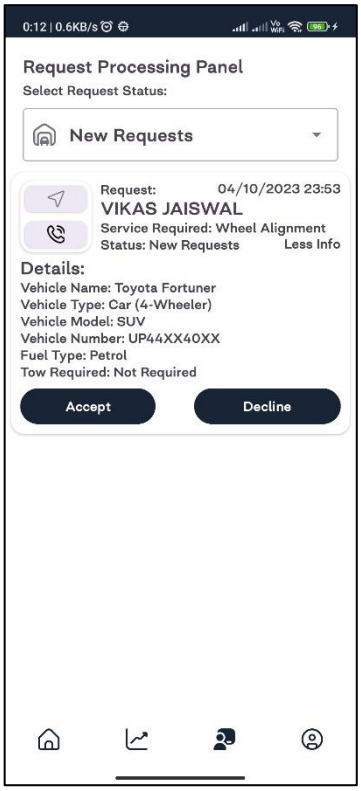
0:08 | 0.0KB/s
Get Directions:
Google Maps
MapBox UI
Garage Showcase Images
Garage Specialization
4-Wheeler
Car Wash
Interior Modification
Car Audio and Electronics
Vehicle Accessories
Ratings Overview
Reviews
Vikas Jaiswal 4.0★ 04/10/2023 23:52
Very nice work at affordable prices
Garage Finder Pro | All Rights Reserved 2023

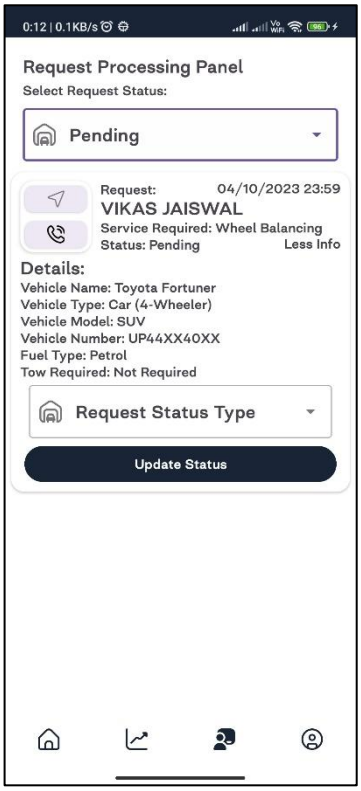

0:27 | 2.4KB/s
Search By Garage Name
New Kurla Garage
Milan Nagar
Janhit Apar
Timing: 09:00 - 17:00
Open
4.0★
Min. Service Cost: ₹250 - ₹500
2-Wheeler
Highly Trained Technicians
Request Service
More Info
Chembur Autom
Milan Nagar
Janhit Apar
Timing: 09:30 - 22:00
Open
4.0★
Min. Service Cost: ₹250 - ₹500
4-Wheeler
Car Wash
Interior Mo
Request Service
Automobile Wash
UP36XX7777
Towing Service Required
*Current Location will be used as the pickup location
Request Service

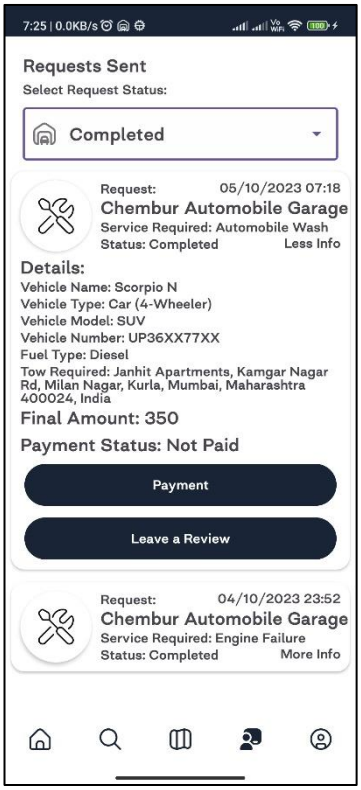

0:10 | 0.1KB/s
Requests Sent
Select Request Status:
New Requests
Request: 04/10/2023 23:52
New Kurla Garage
Service Required: Battery Change
Status: New Requests
Less Info
Details:
Vehicle Name: Scorpio N
Vehicle Type: Car (4-Wheeler)
Vehicle Model: SUV
Vehicle Number: UP36XX77XX
Fuel Type: Diesel
Tow Required: Not Required
Cancel Request

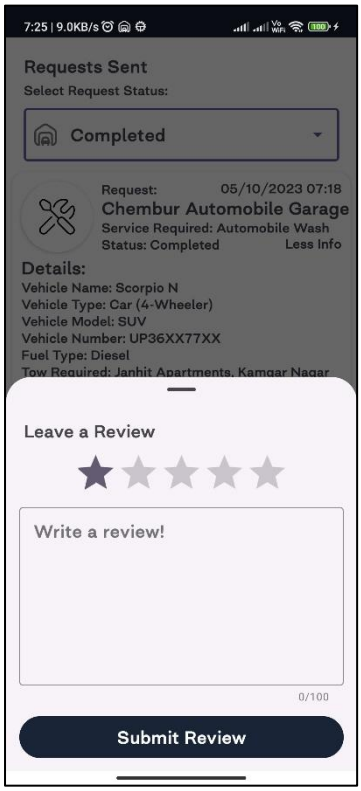

JIO 4G | airtel 0.0KB/s
0:26 Thu, Oct 5
Garage Finder Pro! • now
New Service Request!
You have a new service request from VIKAS JAISWAL
Details:
Service Type: Battery Change
Vehicle Name: Toyota Fortuner Car (4-Wheeler)
Vehicle Number: UP44XX40XX
Fuel Type: Petrol
Vehicle Model: SUV
Tow Service: 1650 Amphitheatre Pkwy, Mountain View, CA 94043, USA
Android System
USB debugging is on Tap to turn off USB d...
USB file transfer is on Tap for more options.

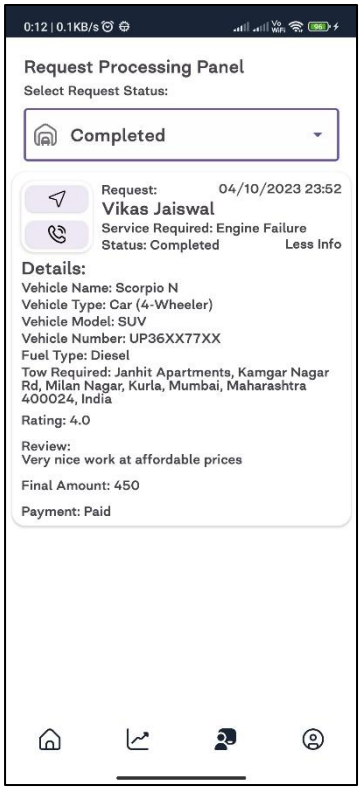

0:12 | 0.3KB/s
Garage Finder Pro!
Welcome!, Chembur Automobile G
Garage Status: Open
Garage Overview
Total Requests: 3
Total Jobcards: 2
Total Earnings: Rs. 450
Job Cards
Pending 1
On Going 0
Completed 1
Pending Jobcards
Request: 04/10/2023 23:53
VIKAS JAISWAL
Wheel Alignment
New Requests
Request: 04/10/2023 23:59
VIKAS JAISWAL
Wheel Balancing
Pending
Garage Finder Pro | All Rights Reserved 2023

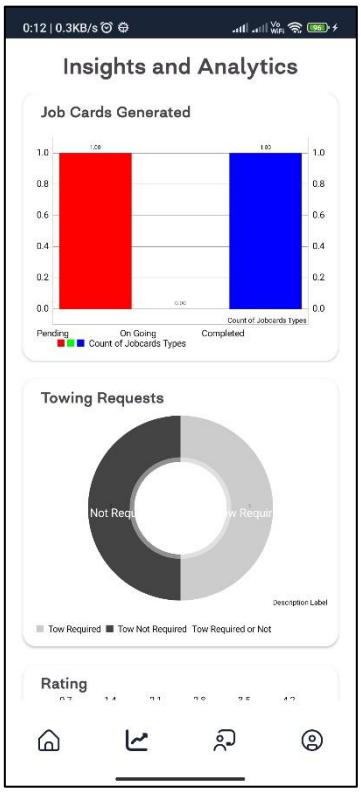

0:12 | 0.6KB/s
Request Processing Panel
Select Request Status:
New Requests
Request: 04/10/2023 23:53
VIKAS JAISWAL
Service Required: Wheel Alignment
Status: New Requests
Less Info
Details:
Vehicle Name: Toyota Fortuner
Vehicle Type: Car (4-Wheeler)
Vehicle Model: SUV
Vehicle Number: UP44XX40XX
Fuel Type: Petrol
Tow Required: Not Required
Accept
Decline


0:12 | 0.1KB/s
Request Processing Panel
Select Request Status:
Pending
Request: 04/10/2023 23:59
VIKAS JAISWAL
Service Required: Wheel Balancing
Status: Pending
Less Info
Details:
Vehicle Name: Toyota Fortuner
Vehicle Type: Car (4-Wheeler)
Vehicle Model: SUV
Vehicle Number: UP44XX40XX
Fuel Type: Petrol
Tow Required: Not Required
Request Status Type
Update Status

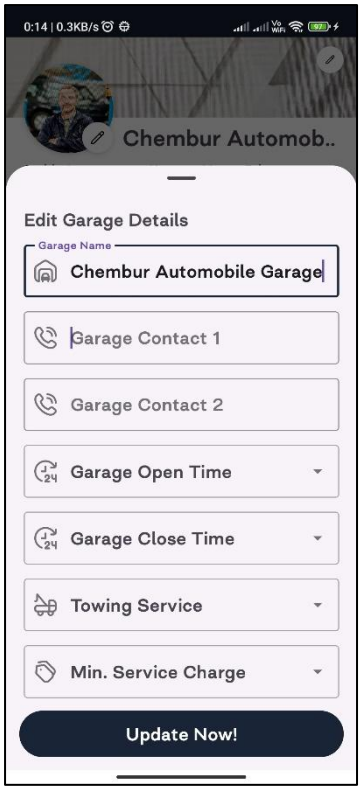

7:25 | 0.0KB/s
Requests Sent
Select Request Status:
Completed
Request: 05/10/2023 07:18
Chembur Automobile Garage
Service Required: Automobile Wash
Status: Completed
Less Info
Details:
Vehicle Name: Scorpio N
Vehicle Type: Car (4-Wheeler)
Vehicle Model: SUV
Vehicle Number: UP36XX77XX
Fuel Type: Diesel
Tow Required: Janhit Apartments, Kamgar Nagar Rd, Milan Nagar, Kurla, Mumbai, Maharashtra 400024, India
Final Amount: 350
Payment Status: Not Paid
Payment
Leave a Review
Request: 04/10/2023 23:52
Chembur Automobile Garage
Service Required: Engine Failure
Status: Completed
More Info

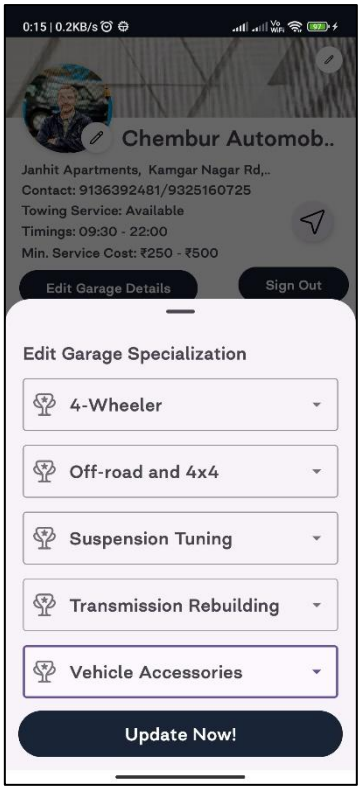

7:25 | 9.0KB/s
Requests Sent
Select Request Status:
Completed
Request: 05/10/2023 07:18
Chembur Automobile Garage
Service Required: Automobile Wash
Status: Completed
Less Info
Details:
Vehicle Name: Scorpio N
Vehicle Type: Car (4-Wheeler)
Vehicle Model: SUV
Vehicle Number: UP36XX77XX
Fuel Type: Diesel
Leave a Review
Write a review!
0/100
Submit Review

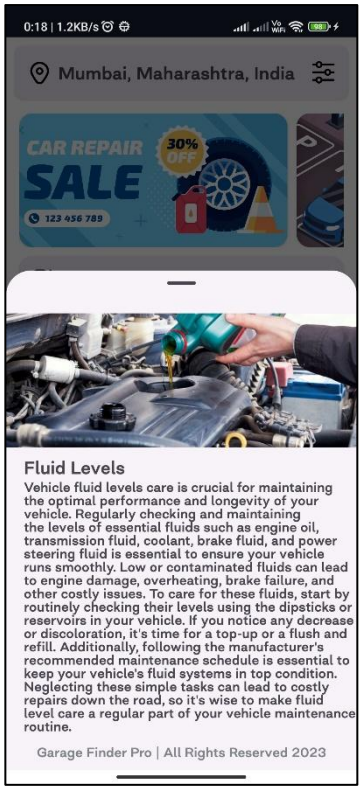

0:12 | 0.1KB/s
Request Processing Panel
Select Request Status:
Completed
Request: 04/10/2023 23:52
Vikas Jaiswal
Service Required: Engine Failure
Status: Completed
Less Info
Details:
Vehicle Name: Scorpio N
Vehicle Type: Car (4-Wheeler)
Vehicle Model: SUV
Vehicle Number: UP36XX77XX
Fuel Type: Diesel
Tow Required: Janhit Apartments, Kamgar Nagar Rd, Milan Nagar, Kurla, Mumbai, Maharashtra 400024, India
Rating: 4.0
Review:
Very nice work at affordable prices
Final Amount: 450
Payment: Paid


0:12 | 0.3KB/s
Insights and Analytics
Job Cards Generated
Pending
On Going
Completed
Count of Jobcards Types
Towing Requests
Tow Required
Tow Not Required
Tow Required or Not
Rating

0:16 | 1.4KB/s
Garage Finder Pro!
We are here to help you!
Name
Email
Message
0/300
Submit
Cancel
Garage Finder Pro | All Rights Reserved 2023


0:14 | 0.3KB/s
Chembur Automob..
Edit Garage Details
Garage Name
Chembur Automobile Garage
Garage Contact 1
Garage Contact 2
Garage Open Time
Garage Close Time
Towing Service
Min. Service Charge
Update Now!


0:15 | 0.2KB/s
Chembur Automob..
Janhit Apartments, Kamgar Nagar Rd,..
Contact: 9136392481/9325160725
Towing Service: Available
Timings: 09:30 - 22:00
Min. Service Cost: ₹250 - ₹500
Edit Garage Details
Sign Out
Edit Garage Specialization
4-Wheeler
Off-road and 4x4
Suspension Tuning
Transmission Rebuilding
Vehicle Accessories
Update Now!


0:18 | 1.2KB/s
Mumbai, Maharashtra, India
CAR REPAIR
SALE
30% OFF
123 456 789
Fluid Levels
Vehicle fluid levels care is crucial for maintaining the optimal performance and longevity of your vehicle. Regularly checking and maintaining the levels of essential fluids such as engine oil, transmission fluid, coolant, brake fluid, and power steering fluid is essential to ensure your vehicle runs smoothly. Low or contaminated fluids can lead to engine damage, overheating, brake failure, and other costly issues. To care for these fluids, start by routinely checking their levels using the dipsticks or reservoirs in your vehicle. If you notice any decrease or discoloration, it's time for a top-up or a flush and refill. Additionally, following the manufacturer's recommended maintenance schedule is essential to keep your vehicle's fluid systems in top condition. Neglecting these simple tasks can lead to costly repairs down the road, so it's wise to make fluid level care a regular part of your vehicle maintenance routine.
Garage Finder Pro | All Rights Reserved 2023

8:03
Garage Finder Pro!
No Internet !
You are Offline!
Garage Finder Pro | All Rights Reserved 2023

Future Enhancements

Advanced Search Filtering:

Advanced search filters for finding the best garages, using Firebase's Firestore database to query and filter results based on criteria like location, ratings, and services offered.

Enhanced Analytics:

Garage owners can have relevant data to track customer behaviour, monitor garage performance, and make data-driven decisions for improvements.

Offline Mode:

Allow users to access basic functionalities even when they have limited or no internet connectivity, ensuring usability in areas with poor network coverage.

Realtime Map View:

Show real-time locations of nearby active garages on a map, integrating Google Maps for live updates and user-friendly navigation.

Garage Management Tools:

Enhance the garage owner dashboard with tools for better business management, such as inventory tracking and employee scheduling.

References

Source: https://medium.com/firebase-developers/android-firebase-authentication-with-google-1c2f6ca3a738

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Source: https://www.geeksforgeeks.org/create-and-add-data-to-firebase-firestore-in-android/

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